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Note

The X-ray crystal structural characterization of dipotassium bisoxalato copper(II) tetrahydrate, $[K_2Cu(ox)_2 \cdot 4H_2O]$ (ox = oxalate dianion)

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Abstract

Dipotassium bisoxalato copper(II) tetrahydrate, $[K_2Cu(ox)_2 \cdot 4H_2O]$ (**1**) (ox = $C_2O_4^{2-}$, oxalate dianion), was first characterized structurally by X-ray crystallography. The compound crystallizes in monoclinic, space group $P2_1/n$ with $a = 3.7770(10)$, $b = 14.819(3)$, $c = 10.756(2)$ Å, $\beta = 93.180(10)°$, $V = 601.1(2)$ Å^3, $Z = 2$, $D_{calc} = 2.154$ g cm^{-3}, $M = 389.84$, $R = 0.0301$ based on 940 observed reflections with $I > 2\sigma(I)$. The results of structural analysis indicate that the complex formed by Cu–O and K–O coordination bonds has a two-dimensional network structure which was further linked by O–H···O hydrogen bonds to give a three-dimensional structure. The Cu(II) atom has square-planar geometry with two bidentate oxalate ligands. Each K^+ cation is coordinated by eight oxygen atoms from one bidentate and two unidentate oxalate groups and four water molecules.

Keywords: Crystal structures; Copper complexes; Oxalato complexes

1. Introduction

The oxalate dianion (ox = $C_2O_4^{2-}$) is known to be versatile upon coordination with metal ions since it can act as a unidentate, chelating or a bridging ligand [1]. On the other hand, the oxalate bridge has been reported to have remarkable ability to transmit electronic effects between paramagnetic centers [2–7]. Therefore the oxalate anion was used widely as a building block to construct frameworks with interesting properties, particularly used in the field of molecular based magnet. Up to now a considerable number of transition metal complexes of oxalate with one-, two- and three-dimensional structures have been reported [8–10]. Among them there have been several X-ray crystallographic studies of copper(II) oxalate dihydrate, dipotassium bisoxalato copper(II) dihydrate, $[K_2Cu(ox)_2 \cdot 2H_2O]$ and dipotassium *trans*-diaquabis(oxalato-*O,O′*)-nickelate(II) tetrahydrate, $[K_2Ni(ox)_2(H_2O)_2 \cdot 4H_2O]$ [8,10,11]. The results indicated that the complex $[Cu(ox) \cdot 2H_2O]$ crystallized in $P2_1/c$ with ribbon-like structure in which the polymeric chains are stacking in a progressively staggered fashion [8,12,13]. However, the structure of dipotassium bisoxalato copper(II) tetrahydrate, $[K_2Cu(ox)_2 \cdot 4H_2O]$ (**1**), is unknown to now, though there are infrared and Raman spectroscopic studies of complex **1** [14]. The structure of $[Cu(ox)_2]^{2-}$ is supposed to be square planar with centrosymmetry since there is no coincidence between the Raman and the infrared bands of this compound. So far no direct evidence concerning the structure of compound **1** was reported.

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We are studying the assembly of imidazole-containing tripodal ligands with transition metal ion [15]. As a new approach self-assembly of tripodal ligands with oxalate metal complexes was carried out in our lab recently. Unfortunately we did not get assemblies from this system; however, we succeeded in obtaining single crystals of dipotassium bisoxalato copper(II) tetrahydrate. In this paper the first X-ray crystallographic study of complex **1** will be described and comparison between the structures of **1** and $[K_2Cu(ox)_2{\cdot}2H_2O]$ (**2**) will also be carried out.

2. Experimental

2.1. Materials

Dipotassium bisoxalato copper(II) dihydrate, $[K_2Cu(ox)_2{\cdot}2H_2O]$ (**2**), was prepared by procedures reported in the literature [10]. All other chemicals are reagent grade and used as received without further purification.

Dipotassium bisoxalato copper(II) tetrahydrate, $[K_2Cu(ox)_2{\cdot}4H_2O]$ (**1**), was obtained as an unexpected products by reaction of **2** with tripodal ligand, 1,3,5-tris(imidazol-1-ylmethyl)-2,4,6-trimethylbenzene (L). A solution of L (36.0 mg, 0.1 mmol) in ethanol (10 ml) was added dropwise to a solution of **2** (106.2 mg, 0.3 mmol) in ethanol (10 ml) and water (20 ml). The undissolved species was removed by filtration after stirring for several hours at room temperature. Dark green crystals of complex **1** were collected after several days by slow evaporation of the clear filtrate; yield: 32%.

Table 1
Crystal data and structure refinement parameters for $[K_2Cu(ox)_2{\cdot}4H_2O]$ (**1**) [a]

Empirical formula	$C_4H_8CuK_2O_{12}$
Formula weight	389.84
Crystal size (mm)	$0.58 \times 0.40 \times 0.40$
Crystal system	monoclinic
Space group	$P2_1/n$
Unit cell dimensions	
a (Å)	3.7770(10)
b (Å)	14.819(3)
c (Å)	10.756(2)
β (°)	93.180(10)
V (Å^3)	601.1(2)
Z	2
D_{calc} (g cm^{-3})	2.154
Absorption coefficient (mm^{-1})	2.569
$F(000)$	390
2θ Range (°)	5.5–50.00
Index ranges	$0 \le h \le 4$, $0 \le k \le 17$, $-12 \le l \le 12$
Reflections collected	1337
Independent reflections	1053 [$R_{int} = 0.0536$]
Data/restraints/parameters	1053/10/105
Goodness-of-fit on F^2	1.078
Final R_1, wR_2 [$I > 2\sigma(I)$]	0.0301, 0.0741
All data	0.0334, 0.0753
Largest difference peak, hole (e Å^{-3})	0.658, −0.863

[a] Weighting scheme: $R = \Sigma||F_o| - |F_c||/\Sigma|F_o|$, $wR_2 = |\Sigma w(|F_o|^2 - |F_c|^2)/\Sigma|w(F_o)^2|^{1/2}$, $w = 1/[\sigma^2(F_o^2) + (0.0521P)^2]$, $P = (F_o^2 + 2F_c^2)/3$.

2.2. Crystal data collection, solution and refinement of the X-ray structure

A single crystal of **1** with dimensions $0.58 \times 0.40 \times 0.40$ mm was mounted and data collection were performed on a Siemens-P4 automatic four-circle diffractometer by ω-scan techniques using graphite-monochromated Mo Kα radiation ($\lambda = 0.71073$ Å) at 294 K. The structure was solved by direct method using SHELXS-97 and refined by full-matrix least-square calculation on F^2 with SHELXL-97 [16]. All non-hydrogen atoms were refined anisotropically, whereas the hydrogen atoms of water molecules were generated geometrically. Calculations were performed on a PC-586 personal computer using Siemens SHELXTL program package [17,18]. Details of the crystal parameters, data collection and refinement are listed in Table 1, and selected bond distances and angles are given in Table 2.

For comparison, the structure of complex **2** was also determined by X-ray crystallography and the data collection was carried out on a Rigaku AFC5R automatic four-circle diffractometer at room temperature. The cell parameters of $a = 8.710(1)$, $b = 9.028(2)$, $c = 6.943(1)$ Å, $\alpha = 99.96(1)$, $\beta = 82.89(1)$, $\gamma = 108.34(1)°$ obtained in this work are similar to those of $a = 8.7092(7)$, $b = 10.3904(5)$, $c = 6.9486(3)$ Å, $\alpha = 121.110(2)$, $\beta = 82.956(2)$, $\gamma = 110.848(2)°$ reported previously with the same space group of $P\bar{1}$ [10].

2.3. Physical measurements

Magnetic measurements on powder sample of complex **1** were carried out using a CHAN-2000 Faraday magnetometer in the 75–300 K temperature range. The apparatus was calibrated with $[Ni(en)_3]S_2O_3$ (en = ethylenediamine). Diamagnetic corrections were made using Pascal's constants.

3. Results and discussion

3.1. Description of the structure

The crystal structures of $[K_2Cu(ox)_2{\cdot}4H_2O]$ (**1**) and $[K_2Cu(ox)_2{\cdot}2H_2O]$ (**2**) indicate that these two compounds have different crystal systems. The title complex **1** crystallizes in the monoclinic form with $P2_1/n$ space group while the dihydrate analog, i.e. complex **2**, crys-

Table 2
Selected bond distances (Å) and bond angles (°) for $[K_2Cu(ox)_2 \cdot 4H_2O]$ (**1**) [a]

Bond distances			
Cu–O(3)	1.929(2)	Cu–O(1)	1.9362(19)
K–O(5)	2.742(2)	K–O(6) # 2	2.797(3)
K–O(4) # 3	2.804(2)	K–O(2) # 4	2.815(2)
K–O(6)	2.827(2)	K–O(2)	2.839(2)
K–O(2) # 5	2.913(2)	K–O(5) # 2	3.049(3)
Bond angles			
O(3)–Cu–O(3) # 1	180.00(9)	O(3)–Cu–O(1)	95.19(8)
O(3) # 1–Cu–O(1)	84.81(8)	O(1)–Cu–O(1) # 1	180.00(12)
O(5)–K–O(6) # 2	115.96(7)	O(5)–K–O(4) # 3	130.35(8)
O(6) # 2–K–O(4) # 3	78.37(7)	O(5)–K–O(2) # 4	96.07(8)
O(6) # 2–K–O(2) # 4	137.36(7)	O(4) # 3–K–O(2) # 4	59.17(6)
O(5)–K–O(6)	68.14(7)	O(6) # 2–K–O(6)	84.38(7)
O(4) # 3–K–O(6)	66.45(7)	O(2) # 4–K–O(6)	82.35(7)
O(5)–K–O(2)	75.28(7)	O(6) # 2–K–O(2)	131.51(7)
O(4) # 3–K–O(2)	132.41(6)	O(2) # 4–K–O(2)	81.59(6)
O(6)–K–O(2)	137.95(7)	O(5)–K–O(2) # 5	152.10(7)
O(6) # 2–K–O(2) # 5	81.14(7)	O(4) # 3–K–O(2) # 5	72.48(7)
O(2) # 4–K–O(2) # 5	82.47(6)	O(6)–K–O(2) # 5	138.39(7)
O(2)–K–O(2) # 5	76.96(7)	O(5)–K–O(5) # 2	81.22(7)
O(6) # 2–K–O(5) # 2	64.36(7)	O(4) # 3–K–O(5) # 2	140.16(7)
O(2) # 4–K–O(5) # 2	153.29(6)	O(6)–K–O(5) # 2	120.17(7)
O(2)–K–O(5) # 2	71.99(6)	O(2) # 5–K–O(5) # 2	87.72(6)

[a] Symmetry transformations used to generate equivalent atoms # 1 $-x-1$, $-y+1$, $-z$; # 2 $x-1$, y, z; # 3 $x+1$, y, $z+1$; # 4 $-x$, $-y+1$, $-z+1$; # 5 $-x-1$, $-y+1$, $-z+1$.

tallized in the triclinic form with the $P\bar{1}$ space group [10]. The crystal structure of complex **1** is shown in Fig. 1(a) with the atom numbering scheme. The asymmetric unit consists of one half of a complex, $[K_2Cu(ox)_2 \cdot 4H_2O]$, with copper(II) atom lying on a center of symmetry. Each copper(II) atom is coordinated by four oxygen atoms of two bidentate oxalate dianions with a square-planar geometry. The copper(II) atom is on the plane composed of O(1), O(3), O(1A) and O(3A) without deviation and the two oxalate units are almost coplanar since the deviation is less than 0.03 Å. The O(3A)–Cu–O(3) and O(1A)–Cu–O(1) angles are both 180.00, which are same as those found in the complex **2**. The other two oxygen atoms of each oxalate group coordinate to the potassium atom which connects the $[Cu(ox)_2]^{2-}$ units (Fig. 1(a)). The dihedral angle between planes containing K(0A), O(2B), K(0B), O(2A) and Cu, O(1), O(3A), O(1A), O(3) is 31.6°. The copper(II) atoms of Cu and Cu(A) is separated by a distance of 11.20 Å. In the structure of complex **2** as exhibited in Fig. 1(b), the repeat unit has two oxalate groups, one copper(II) atom, two potassium atoms and two water molecules. One of the two independent oxalate anions is non-planar since the torsion angles of O(1)–C(1)–C(2)–O(2) and O(3)–C(1)–C(2)–O(4) are 8.7(5) and 7.9(5)°, respectively, while the corresponding torsion angles for oxalate ion containing C(3), C(4) atoms are 2.2(5) and 1.9(6)°. The oxalate ion in sodium and potassium oxalates has been demonstrated by X-ray crystallographic studies to be planar (D_{2h}) [19,20], but for the silver and ammonium salts [21,22] the oxalate ion has a D_2 structure in which the angle between the COO planes is between 22 and 28°.

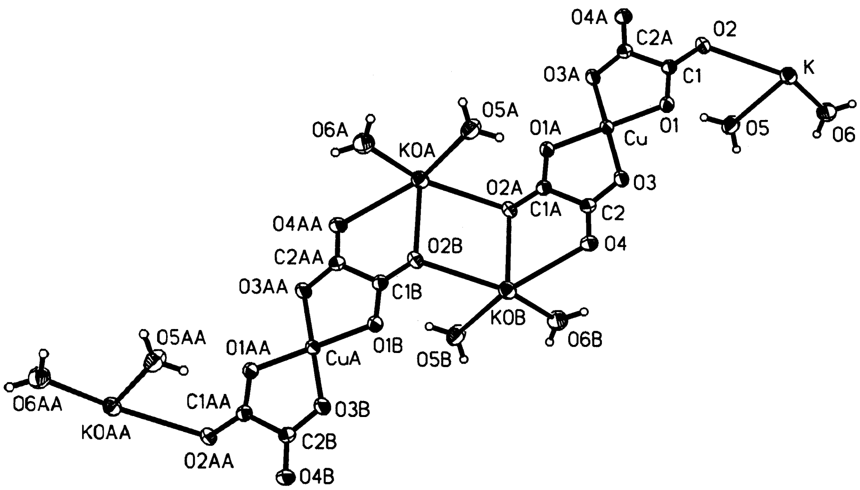
One molecule of $[K_2Cu(ox)_2 \cdot 4H_2O]$ can be repeated and joined by the K–O coordination bond to give a two-dimensional structure for the complex **1** as illustrated in Fig. 2(a). The planes of $[Cu(ox)_2]^{2-}$ unit are parallel each other and the intermetallic distance between the copper(II) atoms, e.g. Cu(A) and Cu(B), is 3.78 Å. Each potassium atom is surrounded by eight oxygen atoms from one bidentate and two unidentate oxalate groups and four water molecules with a strongly distorted square-antiprism geometry. The K–O distances are ranging from 2.742(2) to 3.049(3) Å as listed in Table 2. Similar K–O bond lengths have been observed in the other oxalate metal complexes, for example, $[K_2Ni(ox)_2(H_2O)_2 \cdot 4H_2O]$ has been reported to have K–O distances ranging from 2.779(3) to 3.017(2) Å [11]. The coordination environments around the potassium atoms of complex **1** are shown in Fig. 2(b). The K(0K) and K(0C), K(0C) and K(0M), etc. are bridged by O(2K), O(2C), O(2M) atoms to form two near squares sharing one K(0C)–O(2C) edge since $r_{K(0K)-O(2K)} = r_{K(0C)-O(2C)} = r_{K(0M)-O(2M)} = 2.839(2)$ Å, $r_{K(0K)-O(2C)} = r_{K(0C)-O(2K)} = 2.913(2)$ Å, $r_{K(0M)-O(2C)} = r_{K(0C)-O(2M)} = 2.815(2)$ Å, ∠O(2K)–K(0K)–O(2C) = 76.96(7)°, ∠K(0K)–O(2C)–K(0C) = 103.04(7)°, ∠O(2C)–K(0M)–O(2M) = 81.59(6)°, ∠K(0M)–O(2M)–K(0C) = 98.41(6)°. On the other hand, the K(0K) and K(0M), K(0C) and K(0F), etc. are bridged by two water oxygen atoms O(5M) and O(6M), O(5C) and O(6C), respectively.

The crystal packing diagram of **1** on *bc* plane is illustrated in Fig. 3(a). It can be seen that the network structure shown in Fig. 2(a) was further linked by O–H···O hydrogen bonds (dashed lines) to generate a three-dimensional structure. There are three O(w)–H···O(ox) hydrogen bonds with $r_{O(5)-O(1\,\#\,1)} = 2.851(3)$ Å, ∠O(5)–H–(O1 # 1) = 167(3)°; $r_{O(5)-O(4\,\#\,2)} = 2.826(3)$ Å, ∠O(5)–H–(O4 # 2) = 171(2)°; $r_{O(6)-O(3\,\#\,2)} = 2.888(3)$ Å, ∠O(6)–H–(O3 # 2) = 158(3)° and one O(w)–H···O(w) hydrogen bond with $r_{O(6)-O(5\,\#\,3)} = 3.076(3)$ Å, ∠O(6)–H–(O5 # 3) = 145(3)°, symmetry transformations codes: # 1 $x+1$, y, z; # 2 $x+3/2$, $-y+1/2$, $z+1/2$; # 3 $x+1/2$, $-y+1/2$, $z+1/2$. The nearest distance between two copper(II) atoms of two adjacent layers is 9.29 Å.

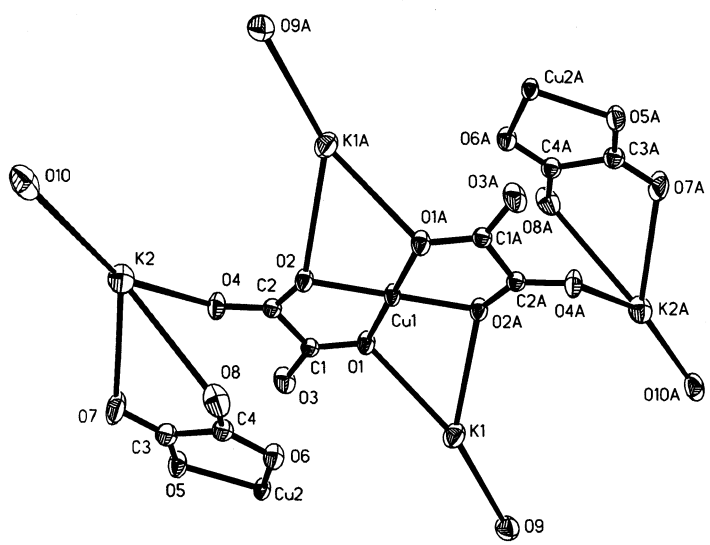
In the case of complex **2**, both K(1) and K(2) atoms are coordinated by eight oxygen atoms, which are from one bidentate, three unidentate and three water molecules and one bidentate, five unidentate and one water molecule, respectively. Such coordination mode makes the complex **2** to have three-dimensional structure as shown in Fig. 3(b). The results demonstrate that the structures of complexes **1** and **2** are quite different although there is no much difference in compositions of **1** and **2**.

3.2. Magnetic properties of the complex

The variable-temperature magnetic susceptibilities of the title complex were measured over the temperature range 70–300 K. The observed magnetic moment is $1.80\mu_B$ ($\mu_B \approx 9.27 \times 10^{-24}$ J T^{-1}) at 300 K and is $1.91\mu_B$ at 75 K per molecule of $[K_2Cu(ox)_2{\cdot}4H_2O]$, which are close to the spin-only value of $1.73\mu_B$ expected for isolated Cu(II) ($S = 1/2$). The slight smooth increase of μ_{eff} with T implies the presence of a very weak ferromagnetic exchange interaction between the nearest Cu(II) ions. In agreement with this the Curie–Weiss plot in the temperature range 75–300 K gives a positive Weiss constant of $\Theta = +11.9$ K (based on

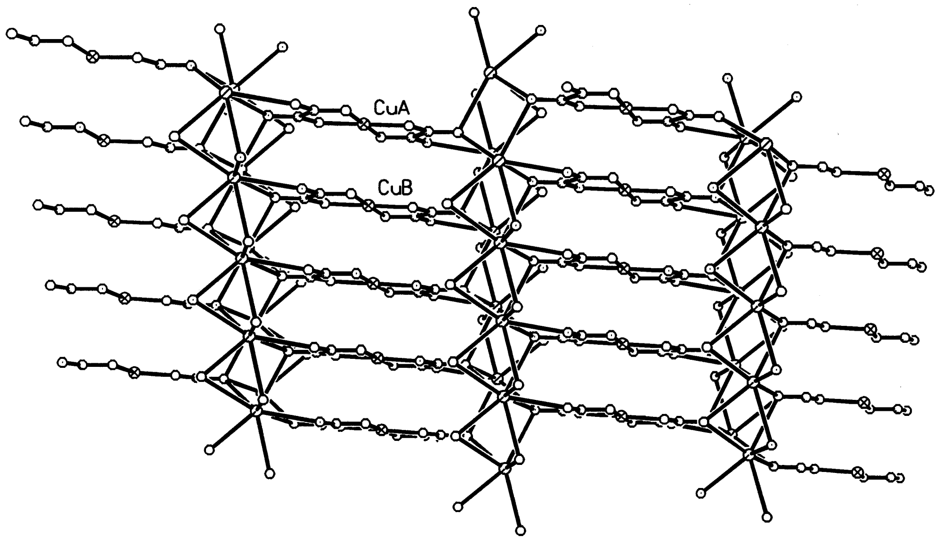


a



b

Fig. 1. (a) Crystal structures of $[K_2Cu(ox)_2{\cdot}4H_2O]$ (**1**) and (b) $[K_2Cu(ox)_2{\cdot}2H_2O]$ (**2**) with atom numbering scheme. The thermal ellipsoids are drawn at 50% probability.



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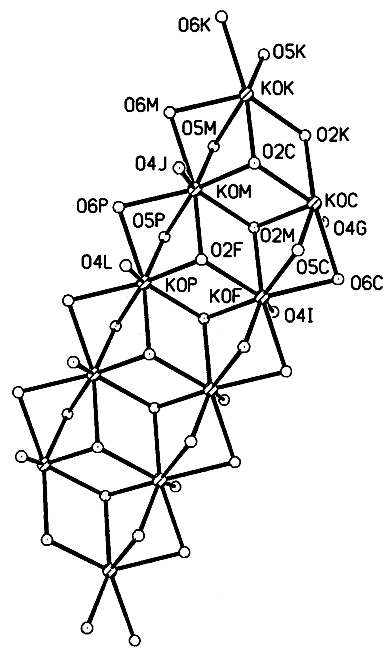
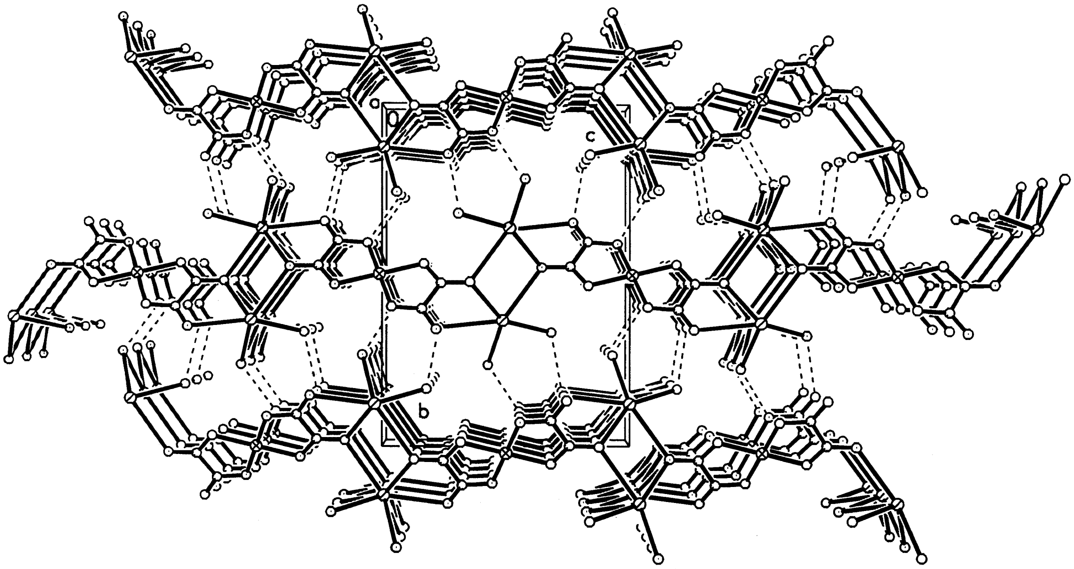


b

Fig. 2. (a) The two-dimensional network of the complex $[K_2Cu(ox)_2 \cdot 4H_2O]$ (**1**) and (b) the coordination environments around the potassium atoms of complex **1**.

$1/\chi_M = C(T-\Theta)$). No strong magnetic interactions were observed due to the long intermetallic distance as revealed by X-ray crystal structure. As shown in Fig. 1(a) the $[Cu(ox)_2]^{2-}$ units in **1** were separated by K–O bonds which prevent the magnetic exchange between the Cu(II) ions.



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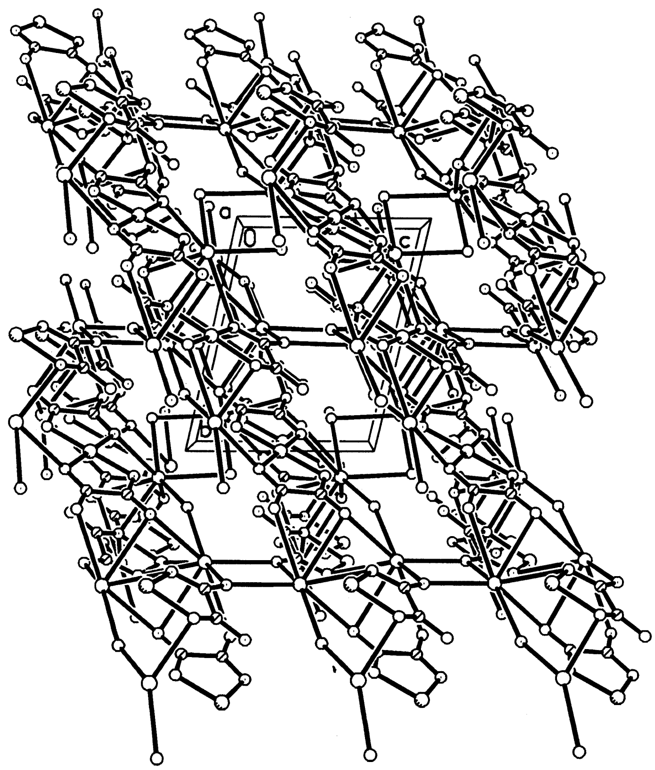

b

Fig. 3. (a) Crystal packing diagram of the complex $[K_2Cu(ox)_2{\cdot}4H_2O]$ (**1**) on *bc* plane with hydrogen bonds indicated by the dashed lines and (b) crystal packing diagram of the complex $[K_2Cu(ox)_2{\cdot}2H_2O]$ (**2**) on *bc* plane.

4. Supplementary material

Crystallographic data for the structural analysis have been deposited with the Cambridge Crystallographic Data Centre, CCDC No. 155318 for compound **1**. Copies of this information may be obtained free of charge from The Director, CCDC, 12 Union Road, Cambridge CB2 1EZ, UK (fax: +44-1223-336-033; e-mail: deposit@ccdc.cam.ac.uk or www: http://www.ccdc.cam.ac.uk).

Acknowledgements

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